PHASE 1 TEAM T072 / SECTION C

METR-LA
Los Angeles

Traffic Speed Forecasting

Independent-Sensor Prediction Using Traditional Machine Learning

FACULTY-ALLOCATED PROJECT

Multi-Agent Traffic Flow Control via Graph Attention Networks and LWR Models



207
Traffic sensors in Los Angeles

34,272
Timestamps (Mar – Jun 2012)

5 minutes
Sampling interval

Predict
15, 30 & 60
minutes ahead

Team Members

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PROJECT REPORT

OCTOBER 2026

METR-LA Dataset & EDA

Data overview and exploratory insights

Sensor geography



207
sensors

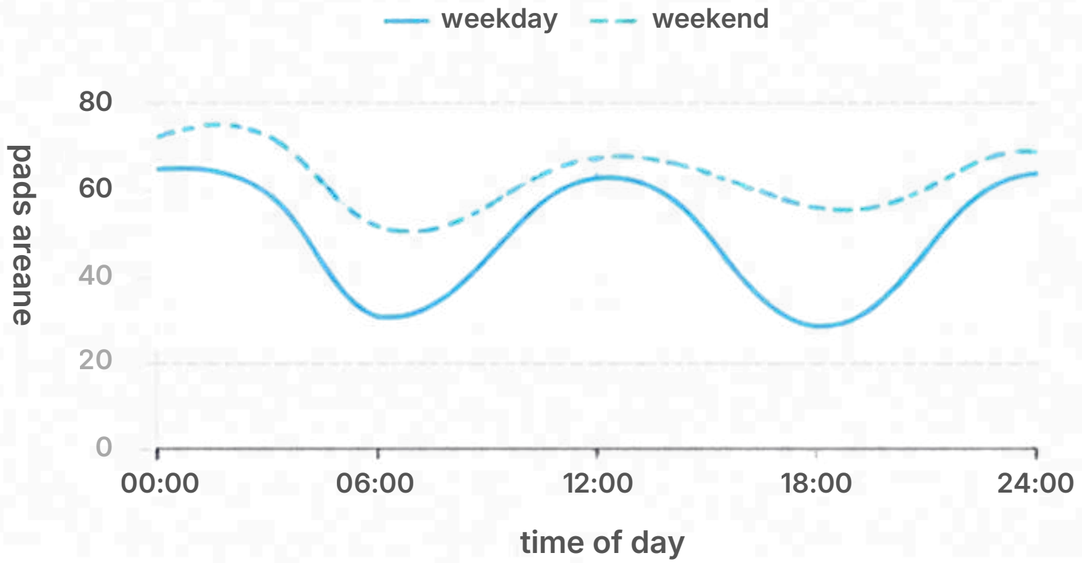
34,272
timestamps

5-minute
cadence

March 1–June 27 2012
data period

8.11%
missing values under
benchmark zero/NaN convention

Average speed profile



Missingness insights

Missingness occurs in contiguous blocks
Aligned with sensor downtime periods
Graph validated for integrity but not used by trained models

Causal Preprocessing & 38 Features

From past readings to predictive features

Past 12 readings
History window from the same sensor

→

Detect missing
Identify gaps within history

→

Forward-fill within history
Fill only using available past values

→

Extract features
Convert history into predictors

Leading missing values are set to the sensor training median. Future targets remain untouched.

12	12	9	5
Lag features	**Missing flags**	**Statistical / change**	**Calendar features**
Historical speed values from the past	Indicators showing where data was absent	Aggregates and change metrics from history	Time-based context from calendar signals

Total features: 38

No standardization, PCA or feature selection ablation.

Four Forecasting Methods

Traditional machine learning approaches for traffic speed prediction

PERSISTENCE

M0

repeats latest
valid reading

approach

uses most recent
observed speed

HISTORICAL AVERAGE

M1

uses training
time-of-week
target slot

approach

averages past values for
same time-of-week

RANDOM FOREST

M3

207 multioutput forests
selected 48 trees
depth 12

key configuration

48 trees
maximum depth 12

XGBOOST

M4

2,484 direct sensor×horizon
regressors
selected 140 trees
depth 3

key configuration

140 trees
maximum depth 3
learning rate 0.06
absoluteerror objective

six
pilot sensors

validation mean absolute error (MAE)

3.829	Random Forest	vs	3.881	Historical Average
3.824	XGBoost	vs	3.413	Historical Average

05 — Chronological Experiment & Evaluation

Data split timeline and evaluation approach



Purge last 11 forecast origins

at earlier boundaries to avoid future-label overlap

Same observed targets used for scoring

MAE	RMSE	MAPE
mean absolute error	root mean squared error	mean absolute percentage error

Test inspected previously | results are exploratory, not a pristine holdout

Results

MAE and RMSE across forecast horizons and models

Model	MAE (mph)			RMSE (mph)		
	15 min	30 min	60 min	15 min	30 min	60 min
Persistence	3.482	4.173	5.321	5.214	6.178	7.645
Historical average	4.187	4.187	4.187	6.021	6.021	6.021
Random Forest	2.972	3.423	3.896	3.982	4.615	5.487
XGBoost	2.920	3.376	3.874	4.046	4.701	5.592

results are exploratory and for comparison only

27.2%

lower 60 min MAE with XGBoost vs persistence

Best RMSE

random forest has the lowest pooled RMSE at all three horizons

07 — Rush Hour & Error Analysis

Mean Absolute Error (mph)

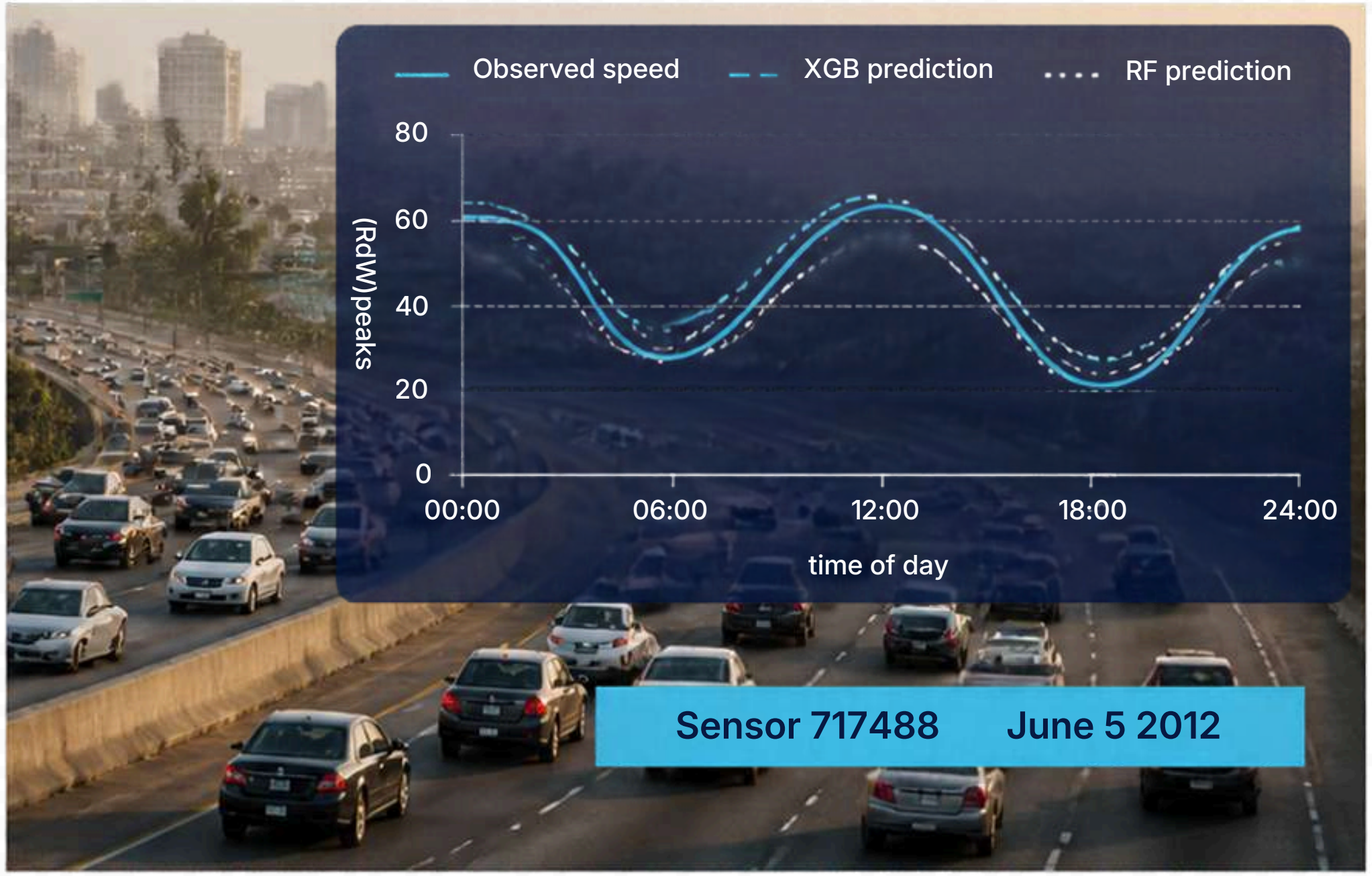
	Random Forest	XGBoost
60-MINUTE PEAK	5.954 mph	6.308 mph
60-MINUTE OFF-PEAK	3.379 mph	3.261 mph

206/207 sensors where XGB beats persistence at 60 minutes

Longer forecasts increase prediction difficulty

Sudden congestion shifts add to forecasting error

Prediction example (Illustrative only)



Conclusions, Limitations & Phase 2

What we established

- Validated speed-only 207-sensor pipeline with 38 causal features
- XGB 3.874 mph 60-min MAE
- RF stronger RMSE and peak MAE

Limitations

- Exposed test
- Limited tuning
- Speed-only
- No neighbour inputs
- Unresolved timezone

Next steps

- Controlled graph-aware GAT experiments
- No-neighbour controls
- LWR needs extra flow density assumptions
- Any control experiment simulated only

3.874 mph | **60-min MAE with XGB**

Thank you / Questions